

Modeling infectious diseases using R

Practical Session:

Transmission within multiple sub populations

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First Training workshop, Cape Town, South Africa
12/10/2026-16/12/2026



Sustainable Training in Epidemiological Analysis and Modelling
for Data-Driven Decision Making in Africa

Part 1: introduction

What do we cover in this practical session ?

- More complicated transmission models in R.
- Models for multiple populations.
- Software: the deSolve package in R.

R program: ModelingIDinR3_V1_MultiPOP_10_2026.R

Interactive tutorial

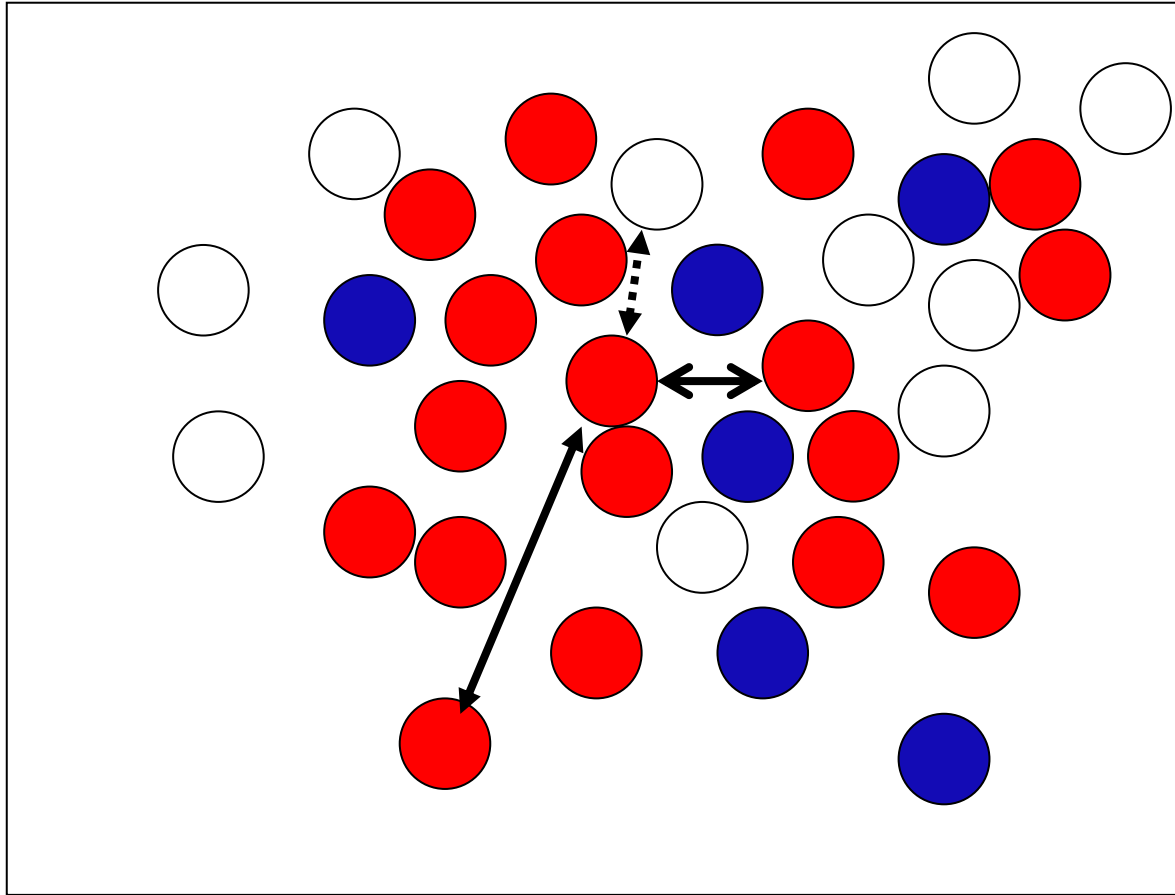
The screenshot shows a web browser window with the following elements:

- Browser Tabs:** "New Tab" and "er_prog3_STEAM3D_2026-V1.Rmd".
- Address Bar:** "File C:/Ziv_Temp_2023/Tarylee_STEAM-3D/Workshop1_12_10_26/ShortCourse/Practicalessions_New/er_prog3_STEAM3D_2026-V1.html".
- Left Sidebar:** A table of contents with three items:
 - 1. The SIR model with multiple populations (highlighted in blue)
 - 2. Example 1: a SIS model for Gonorrhea
 - 3. Example 2: SIR Model With Two Interacting Sub-populations
- Main Content Area:**
 - Date:** 24-09-2026
 - Author:** >eR-BioStat and STAEM-3D
 - Title:** Modeling infectious diseases using R - a multiple populations model
 - Text:** Ziv Shkedy and Rashider Aloni based on Chapter 2-4 in the book *Modeling infectious diseases parameters based on serological and social contact data* (<https://www.simid.be/modeling-infectious-disease-parameters-based-on-serological-and-social-contact-data/>)
 - Section 1:** 1. The SIR model with multiple populations
 - Section 2:** Low of mass action in one population setting
 - Text:** The law of mass action discussed before states that the number of newly infected individuals at time t depends on the number of infected individuals (I), the number of susceptibles (S) and the transmission rate between these two groups (β). In its simplest form, the mass-action principle states:
 - Equation:**
$$\text{Number of new cases} = \beta I \times S = \lambda S.$$
 - Text:** The underlying assumption behind the latter equation is that infected and susceptible individuals mix in a random manner (homogeneous mixing) and therefore β is age and time independent.
 - Section 3:** Multiple populations
 - Text:** In this tutorial we discuss transmission settings in which the social structures in the population determine the mixing patterns in the population. We assume that the population is constructed from multiple sub-populations which may or may not interact with each other. As a result, the transmission process of the infection is governed by the mixing patterns in the population.
 - Section 4:** Heterogenous mixing patterns and Sub-populations
 - Text:** The above assumption about random mixing in the population usually does not hold. Most populations do not mix in a random fashion but are constructed from sub-populations in and between which individuals mix such as for example: different age groups within a school, contacts within households and sexual contacts within the population. Capasso (2008) refers to this mixing pattern in the population as a positive feedback epidemic system. A central characteristic of a positive feedback epidemic system is the mixing or Who Acquires Infection From Whom (WIAIFW) matrix which governs the transmission process. For example, considering a

Rmd program: eR_prog3_STEAM3D_2026_V1.Rmd

Transmission within sub populations

The Mass-Action Principle and the force of infection



Contacts are made in random.

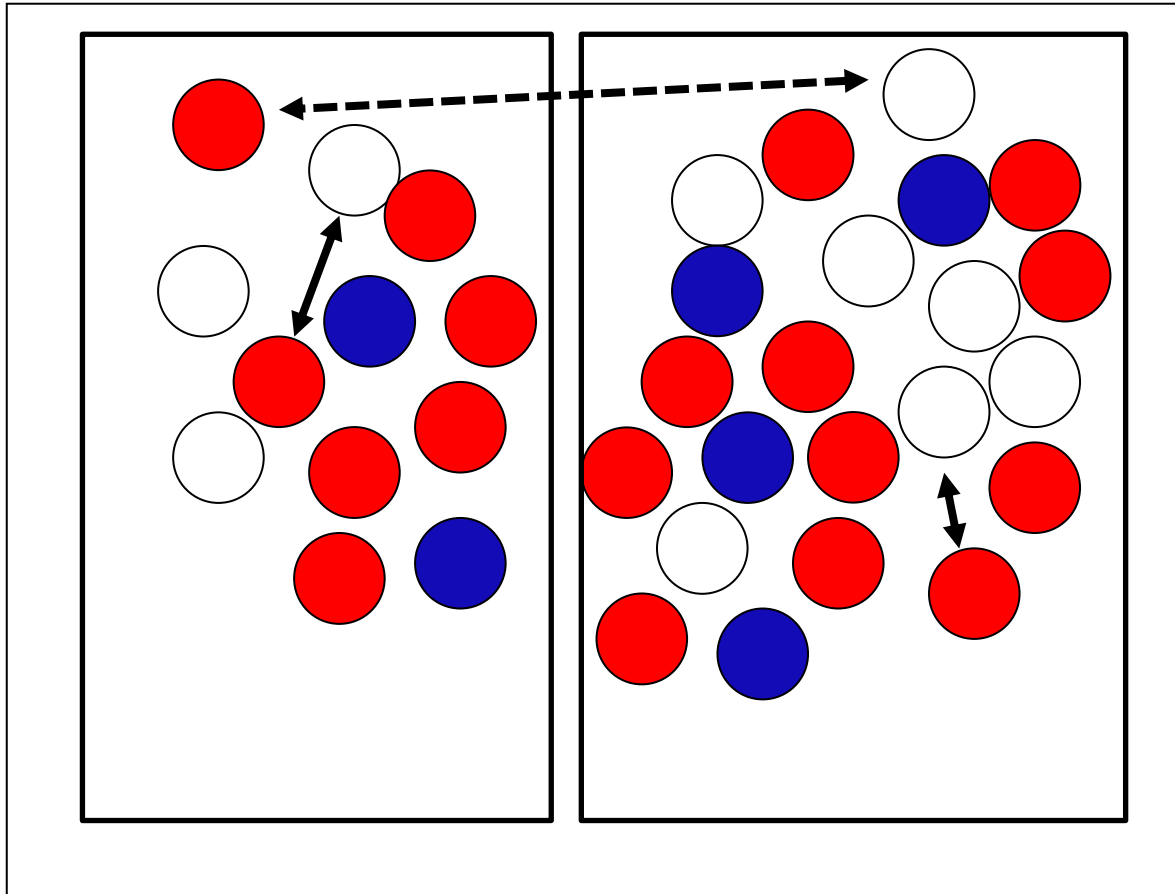
Number of new cases:

$$\beta \times I \times S$$

Transmission
probability per
contact

Number of new cases = $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

Transmission within/between sub populations



Contacts are **NOT** made in random.

Transmission between and within the sub population

Transmission revisited

- Until now, we assumed that transmission is **homogeneous** within the population → unique transmission coefficient β

$$\text{Incidence rate (t)} = \beta \times I(t) \times S(t)$$

- It means that every individual has the same "chance" to be in contact with every other individual within the population
- In general, this is not very realistic!

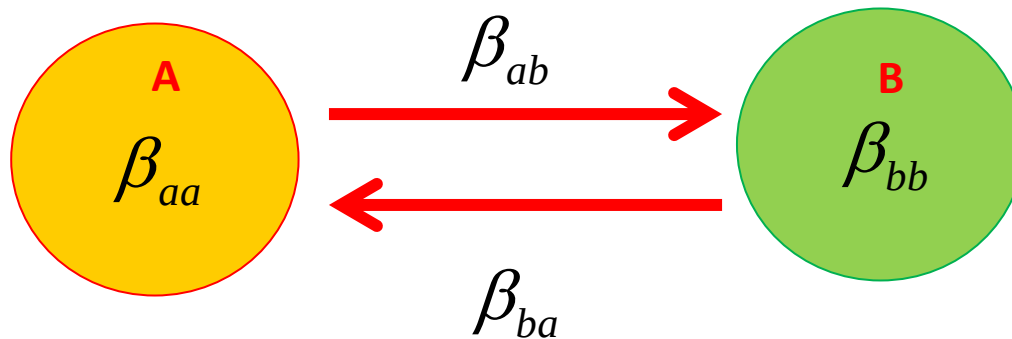
Transmission and mixing pattern

- In general, the “mixing pattern” within the population will **NOT be homogeneous**.
- The population is structured with respect to contacts.
- Examples:
 - Childhood infections: measles, mumps, rotavirus, ...:
greater risk between infants:
in day care centre, school, ..
 - Sexually transmitted infections: contact only possible
within a sexual partnership.

Contact pattern

$$\begin{bmatrix} \beta_{aa} & \beta_{ab} \\ \beta_{ba} & \beta_{bb} \end{bmatrix}$$

Contact matrix

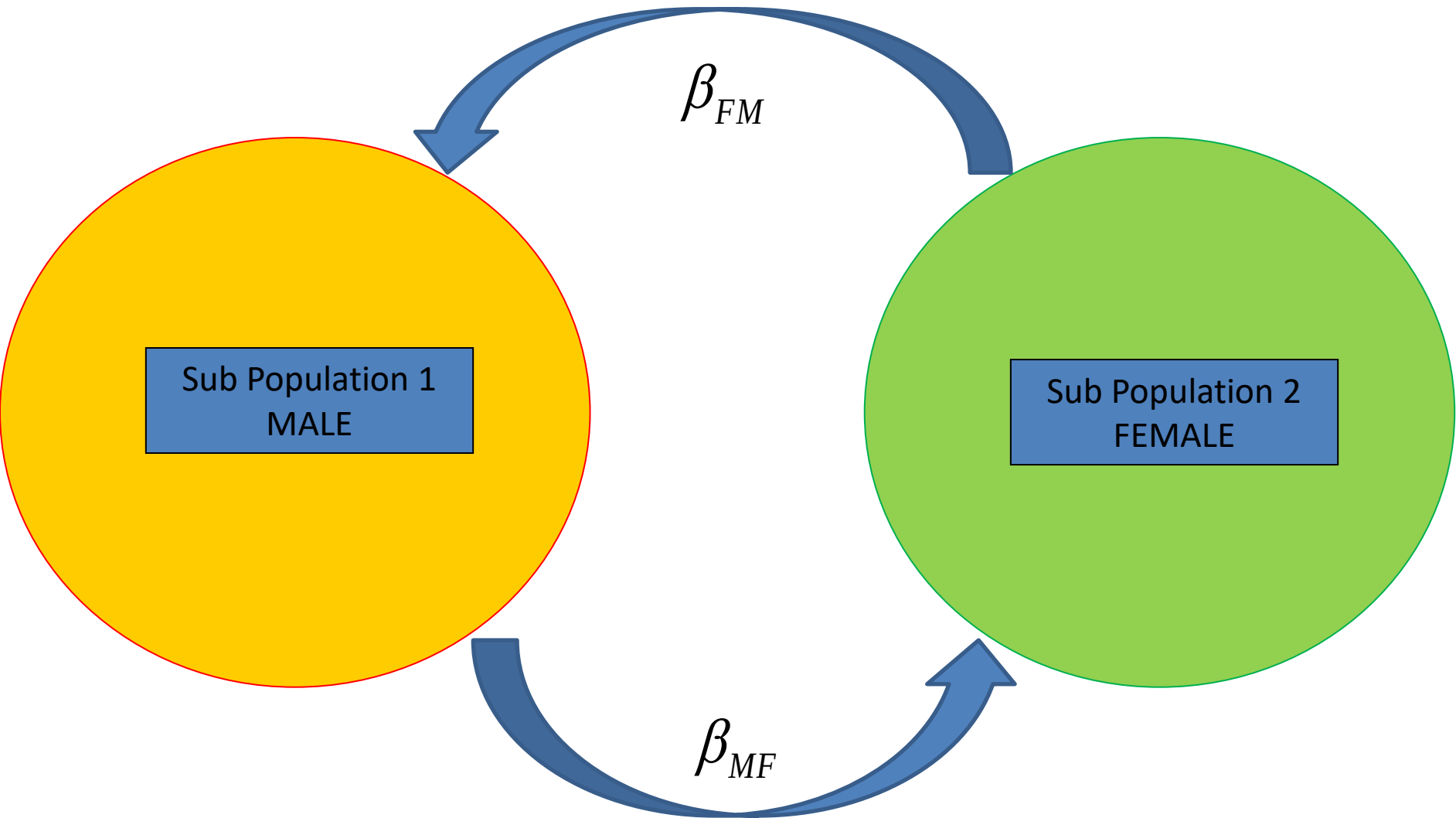


β_{ab} contact between susceptible from sub population A and infected from subpopulation B

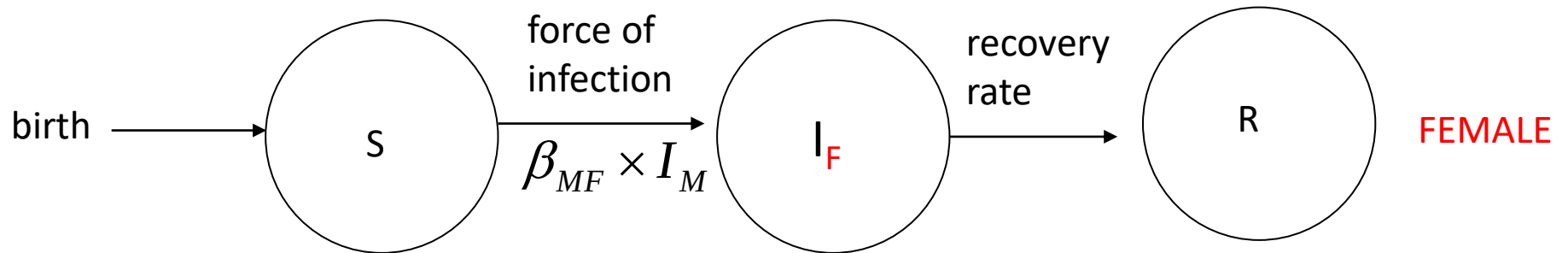
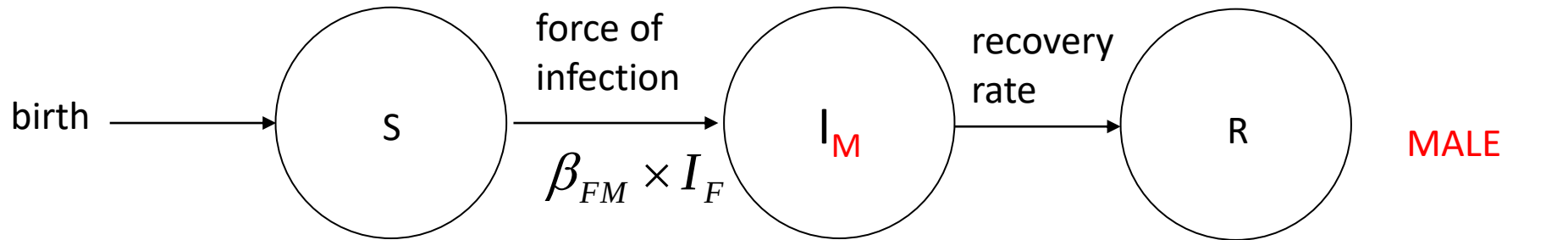
β_{ba} contact between susceptible from sub population B and infected from subpopulation A

Part 2: transmission model for Gonorrhea

Sexually transmission diseases: Gonorrhea

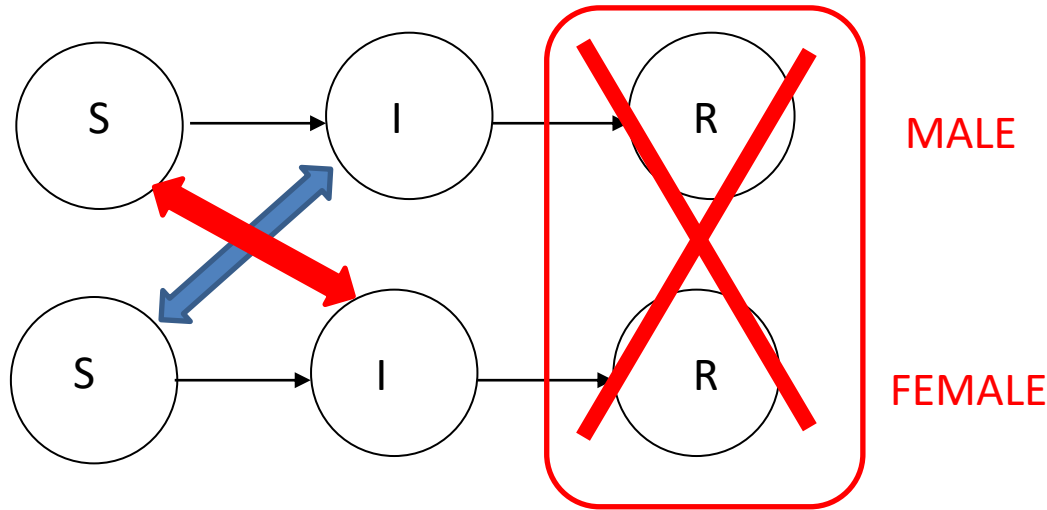


Model structure



$$\begin{bmatrix} \beta_{FM} \\ \beta_{MF} \end{bmatrix} \text{ Contact patterns}$$

Contact pattern: positive feedback

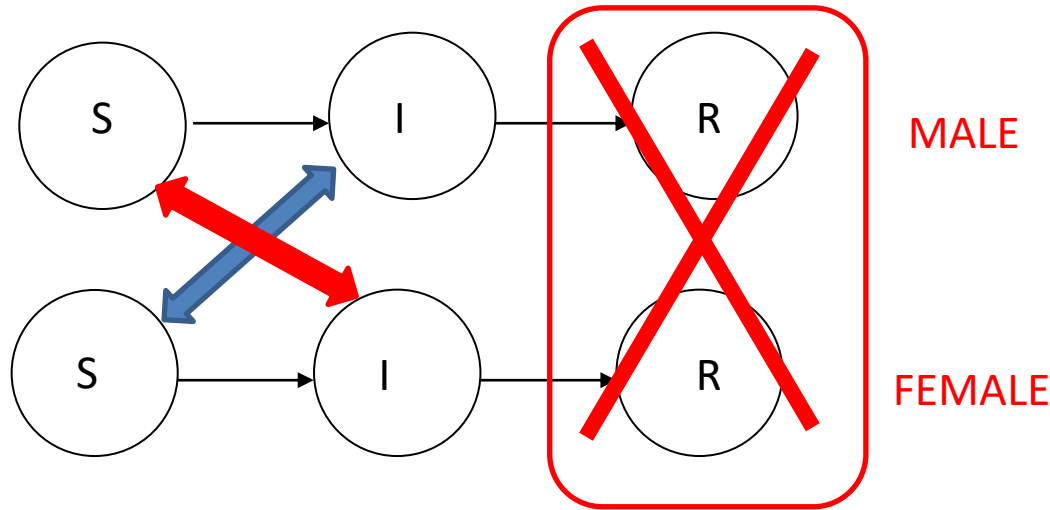


$$\begin{bmatrix} \beta_{MF} & \beta_{FM} \end{bmatrix}$$

The contact matrix represent the interaction between the population of MALE and FEMALE.

Positive feedback: one population influence the transmission of the other population

Transmission model for gonorrhea in R



THE ODE system

$$\frac{dI_F(t)}{dt} = \beta_{MF}(N_F - I_F)I_M - \nu_1 I_F$$
$$\frac{dI_M(t)}{dt} = \beta_{MF}(N_M - I_M)I_F - \nu_2 I_M$$

Contact matrix

```
parameters <- c(beta1=0.000003,beta2=0.000006,  
                v1=0.007,v2=0.05,N1=10000,N2=15000)
```

$$\begin{bmatrix} & \beta_{FM} \\ \beta_{MF} & \end{bmatrix} = \begin{bmatrix} & 0.000003 \\ 0.000006 & \end{bmatrix}$$

Transmission model for gonorrhea in R

```
Gonorrhea<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dY1 <- beta1*(N1-Y1)*Y2-v1*Y1
    dY2 <- beta2*(N2-Y2)*Y1-v2*Y2
    list(c(dY1,dY2))
  })
}
```

$$\frac{dI_F(t)}{dt} = \beta_{MF}(N_F - I_F)I_M - \nu_1 I_F$$
$$\frac{dI_M(t)}{dt} = \beta_{MF}(N_M - I_M)I_F - \nu_2 I_M$$

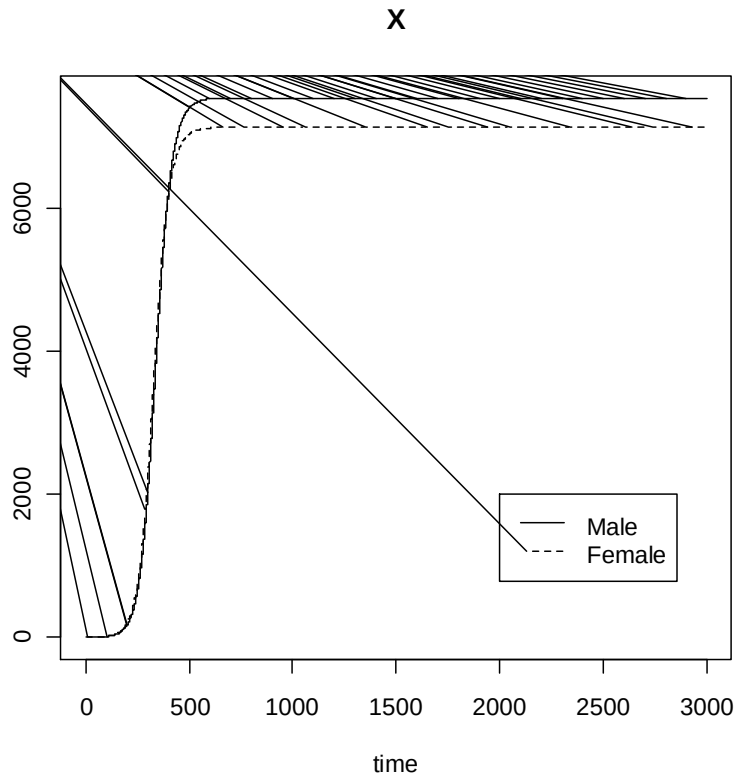
State variables

Population sizes are 10000 & 15000.

At t=one infected in population 1 and zero in population 2.

```
> state <- c(Y1=1, Y2=0)
> state
Y1 Y2
1  0
```

Parameter setting



Parameter setting

$$\beta_{FM} = 0.000003$$

$$\beta_{MF} = 0.000006$$

$$N_M = 10000$$

$$N_F = 15000$$

$$\nu_1 = 0.007$$

$$\nu_5 = 0.05$$

} Different recovery
rate

```
parameters <- c(beta1=0.000003,beta2=0.000006,  
v1=0.007,v2=0.05,N1=10000,N2=15000)
```

The HTML file: example 2

er_prog3_STEAM3D_2026-V1.k... er_prog3_STEAM3D_2026-V1.k... +

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uhasseit.be bookmarks

1. The SIR model with multiple populations

2. Example 1: a SIS model for Gonorrhea

Transmission model for Gonorrhea

The Gonorrhea model in R

Changing the model parameters

Model's output

Changing transmission rates

3. Example 2: SIR Model With Two Interacting Sub-populations

$\beta_{ba} \quad \beta_{bb}$

The mixing rates for individuals from group A with those of group B is denoted as β_{ab} . Individuals can also mix with individuals from the same group. β_{aa} and β_{bb} denote the within-group mixing rates for groups A and B, respectively.

2 Example 1: a SIS model for Gonorrhea

Transmission model for Gonorrhea

Gonorrhea is a disease transmitted through the population by heterosexual contacts. Since immunity to reinfection does not exist, infected individuals, after recovering, move back to the susceptible class. Capasso (2008) proposed a SIS transmission model for Gonorrhea in which the two subpopulations, males and females interact according to the following mixing matrix

$$C = \begin{pmatrix} 0 & \beta_{fm} \\ \beta_{mf} & 0 \end{pmatrix}.$$

The SIS transmission model for gonorrhea can then be described by the following system of (ordinary) differential equations:

$$\begin{cases} \frac{dS_f}{dt} = -\beta_{fm}S_f(t)I_m(t) + v_fI_f(t), \\ \frac{dI_f}{dt} = \beta_{fm}S_f(t)I_m(t) - v_fI_f(t), \\ \frac{dS_m}{dt} = -\beta_{mf}S_m(t)I_f(t) + v_mI_m(t), \\ \frac{dI_m}{dt} = \beta_{mf}S_m(t)I_f(t) - v_mI_m(t). \end{cases}$$

Here, v_f and v_m are the recovery rates for females and males, respectively. Note that the transmission model assumes that mixing between sub-populations occurs in a randomly fashion and therefore the number of new cases equals $\beta_{fm}S_f(t)I_m(t)$ and $\beta_{mf}S_m(t)I_f(t)$ for females and males, respectively. We further assume that $N_i = S_i + I_i$ for both females and males so the SIS model can be reduced to

$$\begin{cases} \frac{dI_f}{dt} = \beta_{fm}(N_f - I_f(t))I_m(t) - v_fI_f(t), \\ \frac{dI_m}{dt} = \beta_{mf}(N_m - I_m(t))I_f(t) - v_mI_m(t). \end{cases}$$

The Gonorrhea model in R

Definition of the model's papameters

For transmission rates $\beta_{fm} = 0.000003$ and $\beta_{mf} = 0.000006$ we have

$$C = \begin{pmatrix} 0 & 0.000003 \\ 0.000006 & 0 \end{pmatrix}.$$

Changing the parameter setting

- Original setting:

$$\begin{bmatrix} & \beta_{FM} \\ \beta_{MF} & \end{bmatrix} = \begin{bmatrix} & 0.00003 \\ 0.000006 & \end{bmatrix}$$

- New setting:

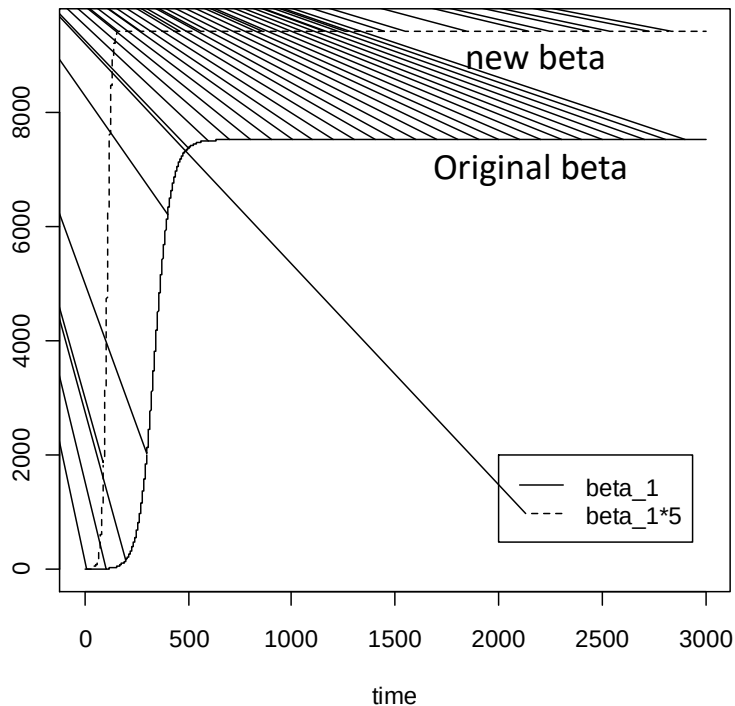
$$\begin{bmatrix} & \beta_{FM} \\ \beta_{MF} & \end{bmatrix} = \begin{bmatrix} & 0.00003 \times 5 \\ 0.000006 & \end{bmatrix}$$

Change in the parameter setting in R

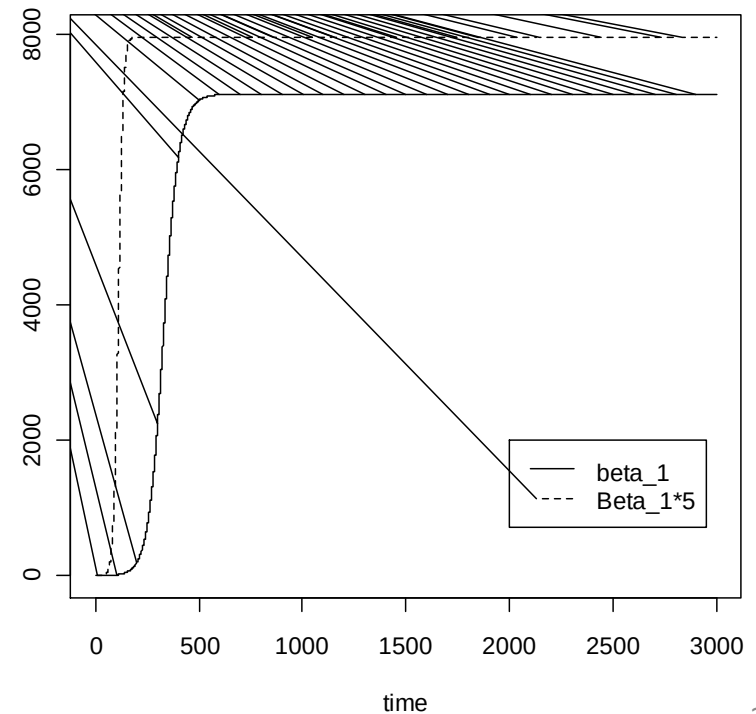
$$\beta_{FM} = \beta_{FM} \times 5$$

```
parameters <- c(beta1=0.000003*5,beta2=0.000006,  
                v1=0.007,v2=0.05,N1=10000,N2=15000)
```

Female



Male



The HTML file

The screenshot shows a web browser window with multiple tabs. The active tab is titled 'er_prog3_STEAM3D_2026-V1.k...' and the address bar shows the file path 'C:/Ziv_Temp_2023/Tarylee_STEAM-3D/Workshop1_12_10_26/ShortCourse/Practicalessons_New/er_prog3_STEAM3D_2026-V1.html'. The browser interface includes a sidebar on the left with a menu of topics, a main content area with text and code blocks, and a Windows taskbar at the bottom.

Sidebar Menu:

- 1. The SIR model with multiple populations
- 2. Example 1: a SIS model for Gonorrhea
 - Transmission model for Gonorrhea
 - The Gonorrhea model in R
 - Changing the model parameters
 - Model's output
 - Changing transmission rates**
- 3. Example 2: SIR Model With Two Interacting Sub-populations

Main Content Area:

time

Figure 2: The Gonorrhea model: number of infected individuals.

Changing transmission rates

A new contact matrix

Next, we investigate the influence of β_{fm} on the model. We run the model with a different mixing matrix in which the parameter β_{mf} is fixed and equal to $\beta_{mf} = 0.000006$ and the parameter β_{fm} increases and equal to 0.000003 . The new contact matrix is given by

$$C = \begin{pmatrix} 0 & 0.000003 \times 5 \\ 0.000006 & 0 \end{pmatrix}.$$

in R, we updated the parameters vector

Show

```
##  beta1  beta2  v1  v2  N1  N2
## 1.5e-05 6.0e-06 7.0e-03 5.0e-02 1.0e+04 1.5e+04
```

while the state vector remains the same.

Show

Model's output

The panels below shows the equilibrium values under the two scenarios. Note that, as before, even though that change in the population was related to the female population, the equilibrium value for the male population was changed due to the positive feedback between the female and male populations. This can also be seen in Figure XXX and CCC.

Show

```
##    time    Y1    Y2
## 30001 3000 7532.051 7121.212
```

Show

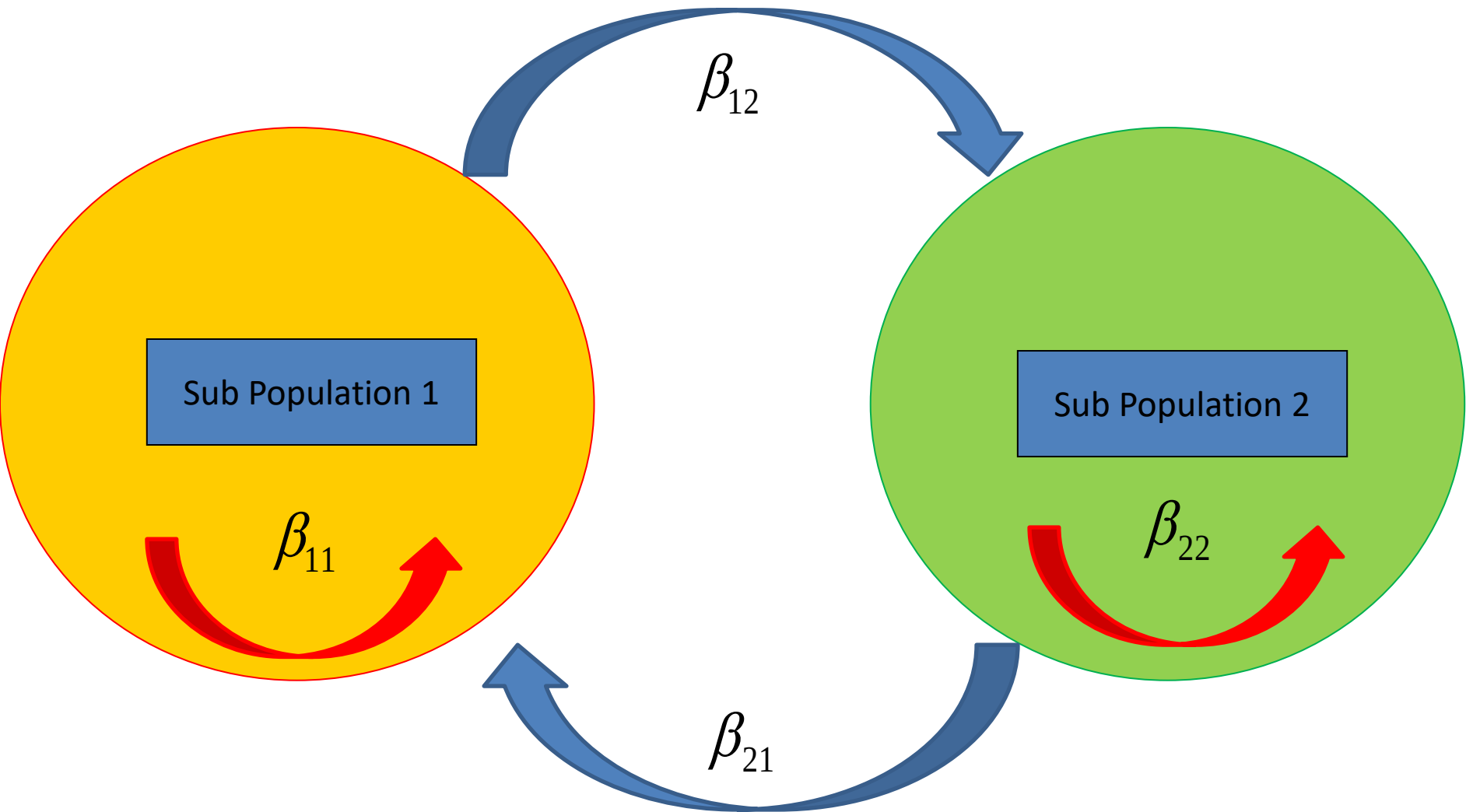
```
##    time    Y1    Y2
## 30001 3000 9446.839 7969.697
```

Show

Female

Part 3: two interacting populations

Two interacting sub population



The HTML file: example 2 (I)

1. The SIR model with multiple populations

2. Example 1: a SIS model for Gonorrhea

3. Example 2: SIR Model With Two Interacting Sub-populations

The mixing matrix

The number of new cases and the force of infection

The SIR model

Implementation in R

Changing the transmission rates

Figure 4: The Gonorrhea model: male population.

3. Example 2: SIR Model With Two Interacting Sub-populations

The mixing matrix

The gonorrhea model discussed in the previous section assumes that there is no mixing within each subpopulation ($\beta_{mm} = \beta_{ff} = 0$). In this section we consider a SIR model in which individuals are assumed to make contact within and across sub-populations. For a case with two sub-populations the WAIFW or mixing matrix is given by

$$C = \begin{pmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{pmatrix}.$$

For each sub-population, the population size is $N_i = S_i + I_i + R_i$, $i = 1, 2$.

The number of new cases and the force of infection

The main difference between the current model to the SIR model for a single population is that in the current model the number of new cases depends on the mixing pattern within and across sub-populations, that is

$$\left(\sum_{j=1}^2 \beta_{ij} I_j \right) S_i = \lambda_i S_i.$$

The SIR model

The differential equation system for the i th subpopulation is given by

$$\begin{cases} \frac{dS_i(t)}{dt} = - \left(\sum_{j=1}^2 \beta_{ij} I_j \right) S_i + N_i \mu_i - \mu_i S_i, \\ \frac{dI_i(t)}{dt} = \left(\sum_{j=1}^2 \beta_{ij} I_j \right) S_i - (\mu_i + \nu_i) I_i, \\ \frac{dR_i(t)}{dt} = \nu_i I_i - \mu_i R_i. \end{cases}$$

Implementation in R

Let us consider the special case of symmetric mixing matrix in the population given by

Contact pattern

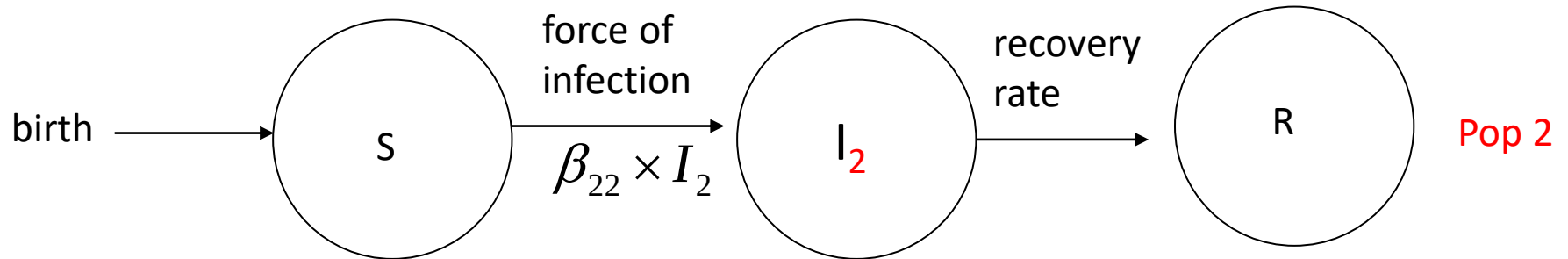
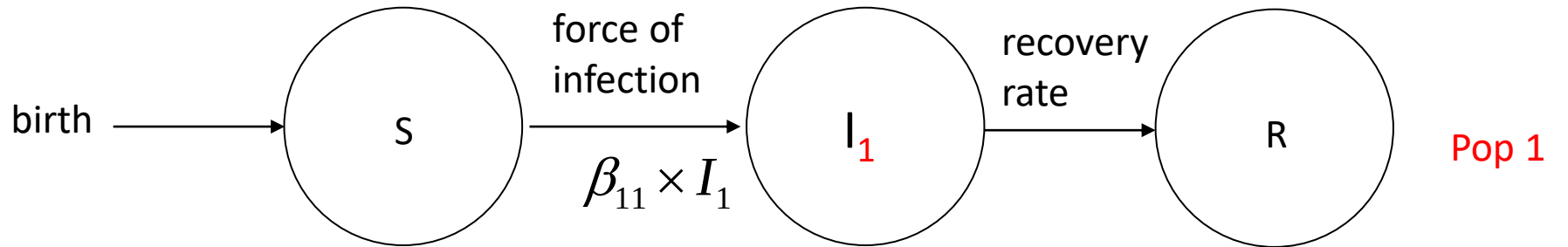
$$\begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{12} & \beta_{22} \end{bmatrix}$$

The contact matrix represent the interaction between the two sub populations.

Within population and within and between transmission.

```
parameters <- c(beta11=0.00,beta12=0.075,beta21=0.075,beta22=0.05,  
v1=1/13,v2=1/30,mu=0.001)
```

Model structure 1

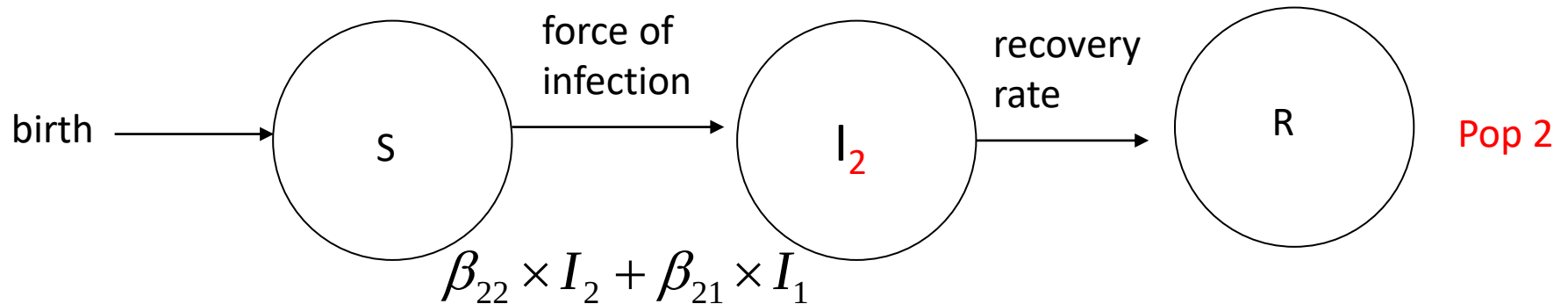
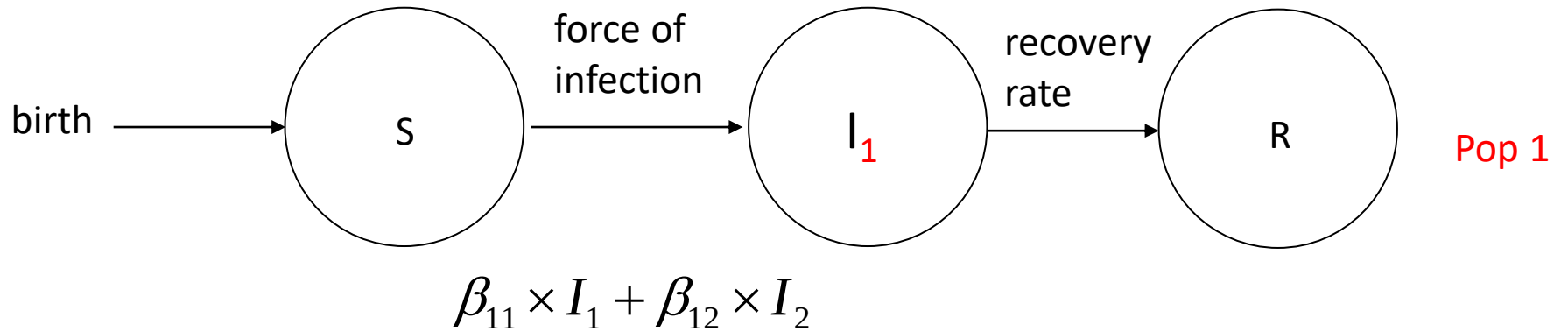


$$\begin{bmatrix} \beta_{11} & 0 \\ 0 & \beta_{22} \end{bmatrix}$$

Contact patterns: only within group transmission

```
parameters <- c(beta11=0.05,beta12=0.000,beta21=0.000,beta22=0.05,  
v1=1/13,v2=1/30,mu=0.001)
```

Model structure 2



$$\begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}$$

Contact patterns: within/between transmission

```
parameters <-  
c(beta11=0.00,beta12=0.075,beta21=0.075,beta22=0.05,  
  v1=1/13,v2=1/30,mu=0.001)
```

Model Structure

$$\begin{aligned}\frac{dS_1(t)}{dt} &= \mu - (\beta_{11}I_1 + \beta_{12}I_2)S_1 - \mu S_1 \\ \frac{dI_1(t)}{dt} &= (\beta_{11}I_1 + \beta_{12}I_2)S_1 - \nu_1 I_1 - \mu I_1 \\ \frac{dR_1(t)}{dt} &= \nu_1 I_1 - \mu R_1 \\ \frac{dS_2(t)}{dt} &= \mu - (\beta_{21}I_1 + \beta_{22}I_2)S_2 - \mu S_2 \\ \frac{dI_2(t)}{dt} &= (\beta_{21}I_1 + \beta_{22}I_2)S_2 - \nu_2 I_2 - \mu I_2 \\ \frac{dR_2(t)}{dt} &= \nu_2 I_2 - \mu R_2\end{aligned}$$

Structure in R

```
SIRtwo<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dY1 <- -(beta11*Y2+beta12*Y5)*Y1+mu-mu*Y1
    dY2 <- (beta11*Y2+beta12*Y5)*Y1-v1*Y2-mu*Y2
    dY3 <- v1*Y2 - mu*Y3
    dY4 <- -(beta21*Y2+beta22*Y5)*Y4+mu-mu*Y4
    dY5 <- (beta21*Y2+beta22*Y5)*Y4-v2*Y5-mu*Y5
    dY6 <- v2*Y5-mu*Y6
    list(c(dY1,dY2,dY3,dY4,dY5,dY6))
  })
}
```

State variables

We run the model in a “proportion” setting (population size=1).

At $t = 0$, 80% are susceptible and 20% are infected (at each sub population)

```
> state <- c(Y1=0.8,Y2=0.2,Y3=0,Y4=0.8,Y5=0.2,Y6=0)
> state
  Y1  Y2  Y3  Y4  Y5  Y6
0.8 0.2 0.0 0.8 0.2 0.0
```

What do we do now ?

- We run the model twice, with two parameter setting:

Parameter setting 1



$$\begin{bmatrix} \beta_{11} = 0.05 & \beta_{12} = 0.05 \\ \beta_{12} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$

Parameter setting 2



$$\begin{bmatrix} \beta_{11} = 0.05 & \beta_{12} = 0.075 \\ \beta_{12} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$

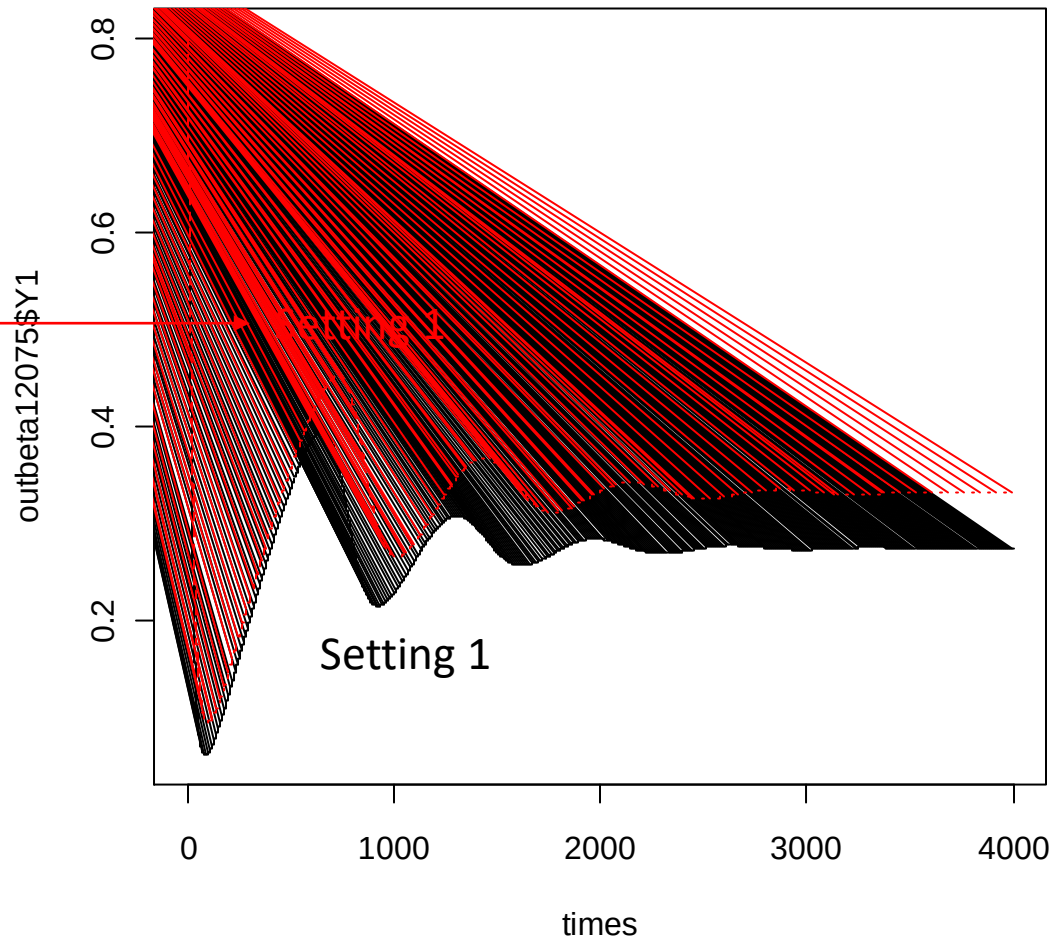
Example

Susceptible in population 1

Parameter setting

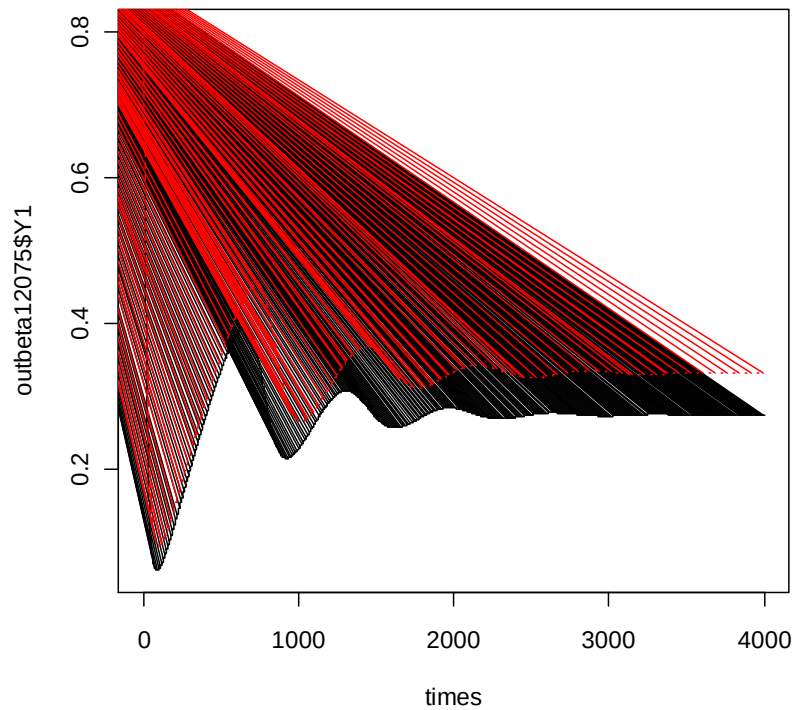
$$\begin{bmatrix} \beta_{11} = 0.05 & \beta_{12} = 0.075 \\ \beta_{12} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$

$$\begin{bmatrix} \beta_{11} = 0.05 & \beta_{12} = 0.05 \\ \beta_{12} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$

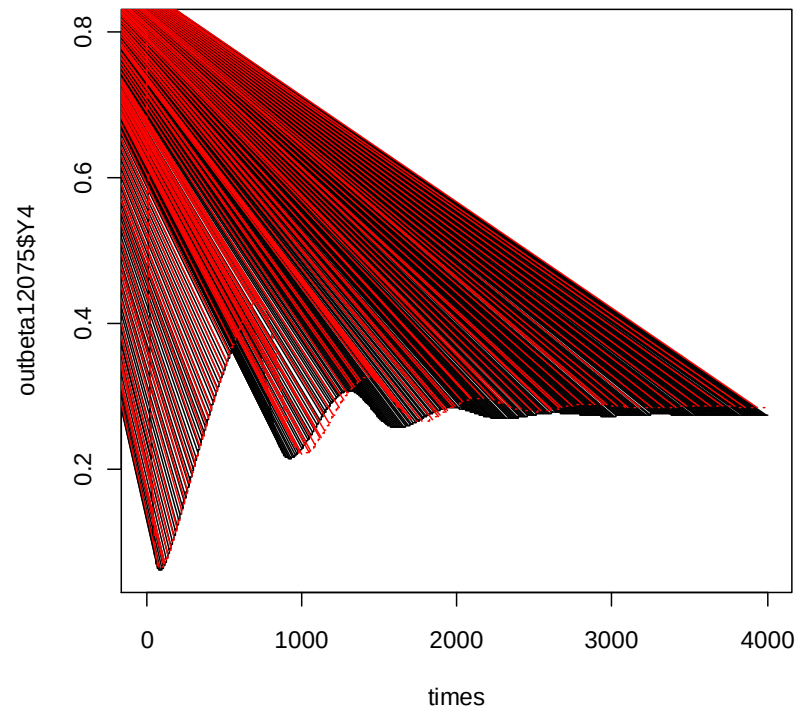


Susceptible in the two sub population

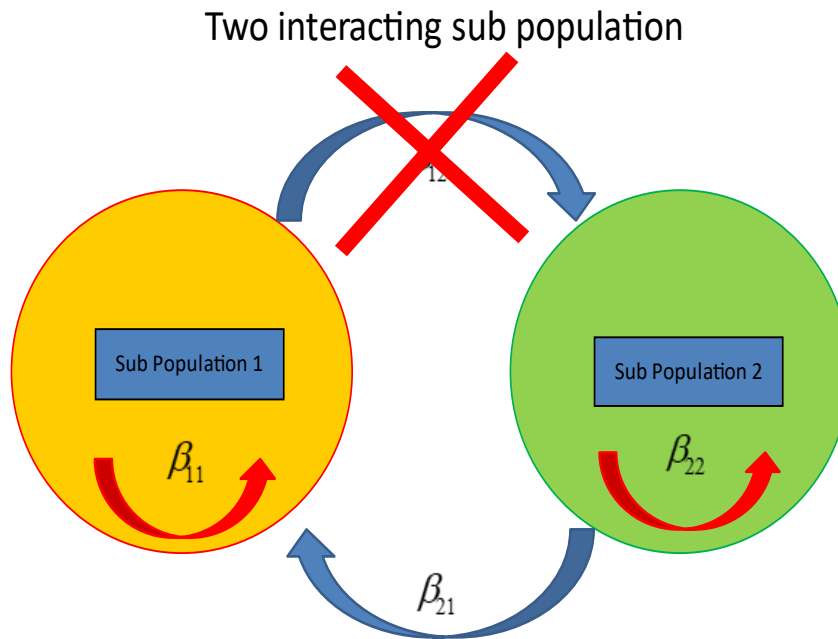
Susceptible in population 1



Susceptible in population 2



Changing the parameter setting (II)



Transmission from population 1 to population 2 is not possible.

The HTML file: example 2 (II)

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er_prog3_STEAM3D_2026-V1.html Open in Browser Find Publish

1. The SIR model with multiple populations

2. Example 1: a SIS model for Gonorrhea

3. Example 2: SIR Model With Two Interacting Sub-populations

The mixing matrix

The number of new cases and the force of infection

The SIR model

Implementation in R

Changing the transmission rates (I)

Changing the transmission rates (II)

Changing the transmission rates (II)

The new contact matrix with $\beta_{12} = 0.0$

In this example, $\beta_{12} = 0$. Hence, the contact matrix is given by

$$C = \begin{pmatrix} 0.05 & 0.0 \\ 0.075 & 0.05 \end{pmatrix}.$$

Show

##	beta11	beta12	beta21	beta22	v1	v2	mu
##	0.05000000	0.00000000	0.07500000	0.05000000	0.03333333	0.03333333	0.00100000

Running the model

Show

##	time	Y1	Y2	Y3	Y4	Y5	Y6	
##	1	0.00	0.80000000	0.20000000	0.000000e+00	0.80000000	0.20000000	0.000000e+00
##	2	0.01	0.7999220	0.2000113	6.666822e-05	0.7998020	0.2001313	6.668822e-05
##	3	0.02	0.7998440	0.2000227	1.333396e-04	0.7996040	0.2002626	1.334195e-04
##	4	0.03	0.7997660	0.2000340	2.000140e-04	0.7994060	0.2003938	2.001939e-04
##	5	0.04	0.7996880	0.2000453	2.666915e-04	0.7992079	0.2005251	2.670114e-04
##	6	0.05	0.7996101	0.2000566	3.333722e-04	0.7990099	0.2006562	3.338720e-04

Show

Model's output

Show

18°C Bewölkt 9:36 1/10/2026

36

Example

Parameter setting

$$\begin{bmatrix} \beta_{11} = 0.05 & \beta_{12} = 0.075 \\ \beta_{21} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$

```
parameters <-  
c(beta11=0.05,beta12=0.075,beta21=0.075,beta22=0.05,  
  v1=1/13,v2=1/30,mu=0.001)
```

$$\begin{bmatrix} \beta_{11} = 0.05 & \beta_{12} = 0.00 \\ \beta_{21} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$

```
parameters <-  
c(beta11=0.05,beta12=0.00,beta21=0.075,beta22=0.05,  
  v1=1/13,v2=1/30,mu=0.001)
```

In the second setting:

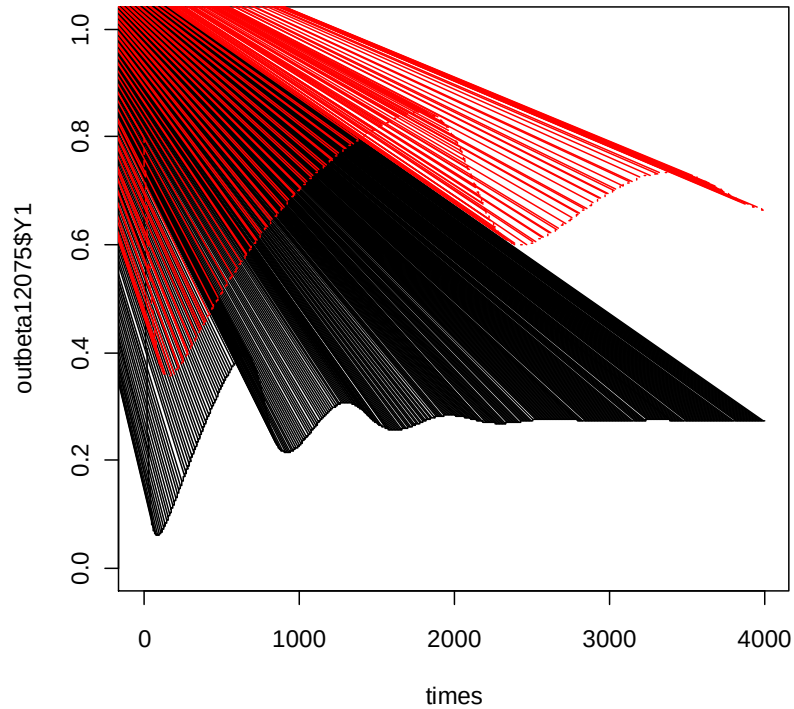
Per contact:

Susceptible from pop. 1 with infected pop. 2: no transmission.

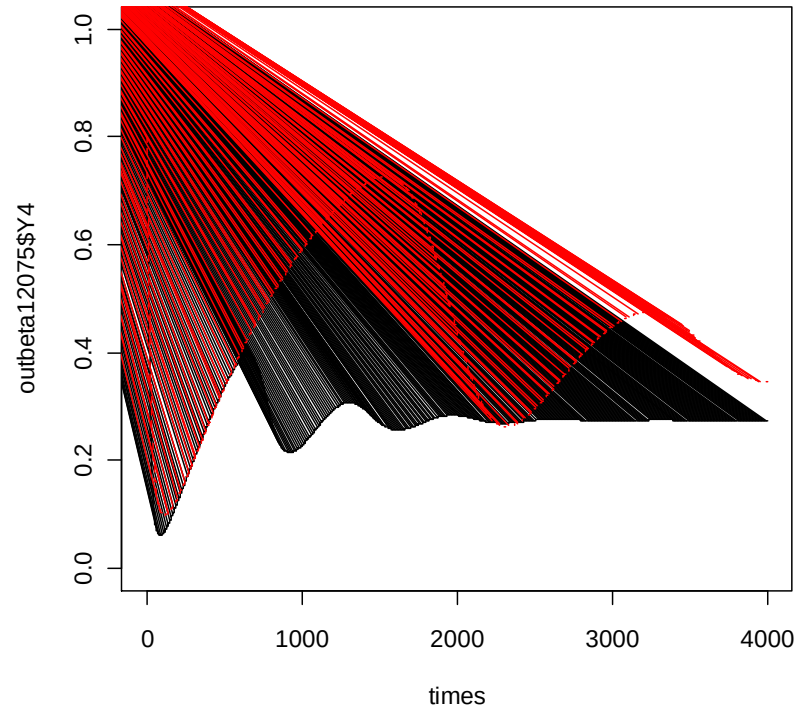
Susceptible from pop. 2 with infected pop. 1: transmission is possible.

Susceptible in the two sub population

Susceptible in population 1

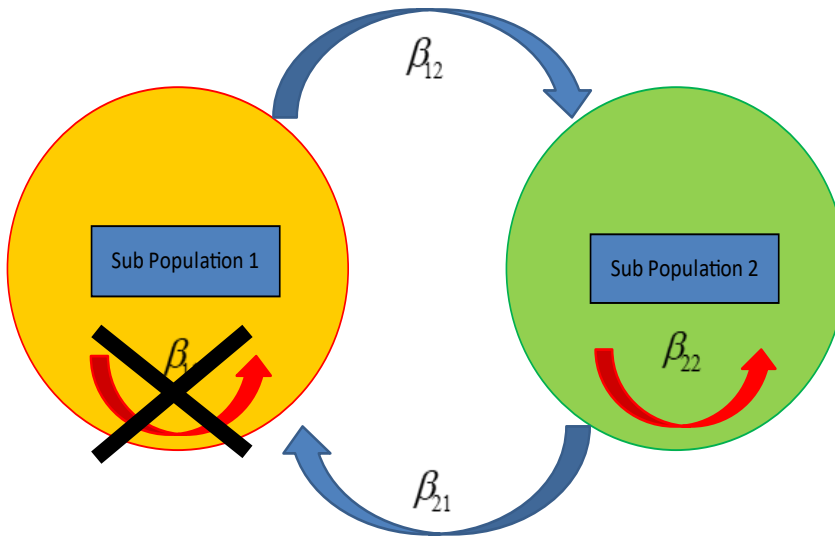


Susceptible in population 2



Changing the parameter setting (2)

Two interacting sub population



Transmission with population 1 is not possible.

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The HTML file: example 2 (III)

Figure 9: The Gonorrhea model.

Changing the transmiison rates (III)

The new contact matrix with $\beta_{12} = 0.0$

In thi example, $\beta_{11} = 0$. Hence, the contact metrix is given by

$$C = \begin{pmatrix} 0.0 & 0.007 \\ 0.075 & 0.05 \end{pmatrix}.$$

beta11 beta12 beta21 beta22 v1 v2 mu

0.00000000 0.07500000 0.07500000 0.05000000 0.03333333 0.03333333 0.00100000

Running the model

time Y1 Y2 Y3 Y4 Y5 Y6

1 0.00 0.8000000 0.2000000 0.000000e+00 0.8000000 0.2000000 0.000000e+00

2 0.01 0.7998820 0.2000514 6.667489e-05 0.7998020 0.2001313 6.668822e-05

3 0.02 0.7997639 0.2001028 1.333662e-04 0.7996039 0.2002626 1.334195e-04

4 0.03 0.7996457 0.2001542 2.000741e-04 0.7994059 0.2003939 2.001940e-04

5 0.04 0.7995275 0.2002057 2.667984e-04 0.7992077 0.2005252 2.670115e-04

6 0.05 0.7994093 0.2002572 3.335392e-04 0.7990096 0.2006565 3.338721e-04

Model's output

Example: positive feedback

Parameter setting

$$\begin{bmatrix} \beta_{11} = 0.05 & \beta_{12} = 0.075 \\ \beta_{21} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$



$$\begin{bmatrix} \beta_{11} = 0.0 & \beta_{12} = 0.075 \\ \beta_{21} = 0.075 & \beta_{22} = 0.05 \end{bmatrix}$$

In the second setting:

First population:

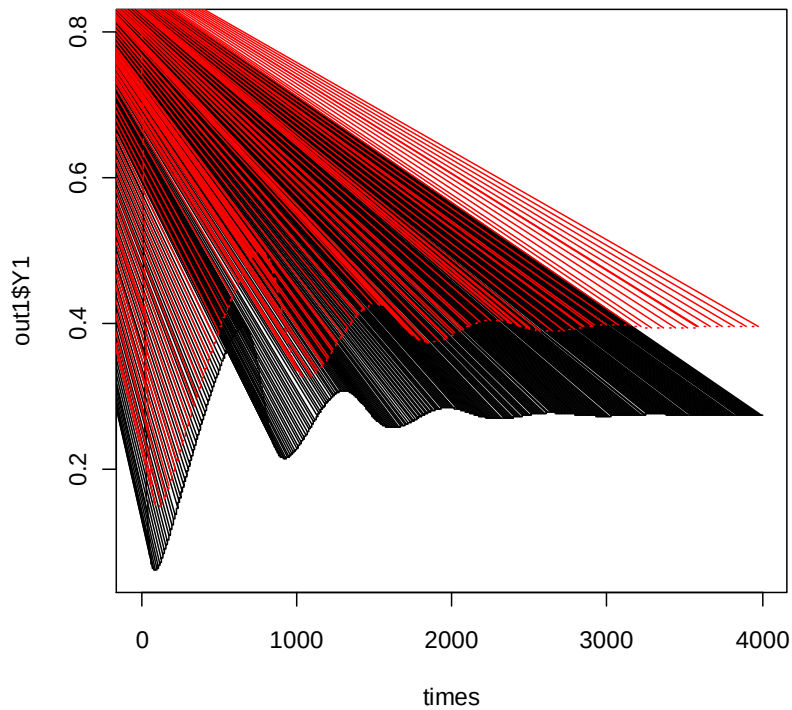
All transmission via contact between the population

Second population:

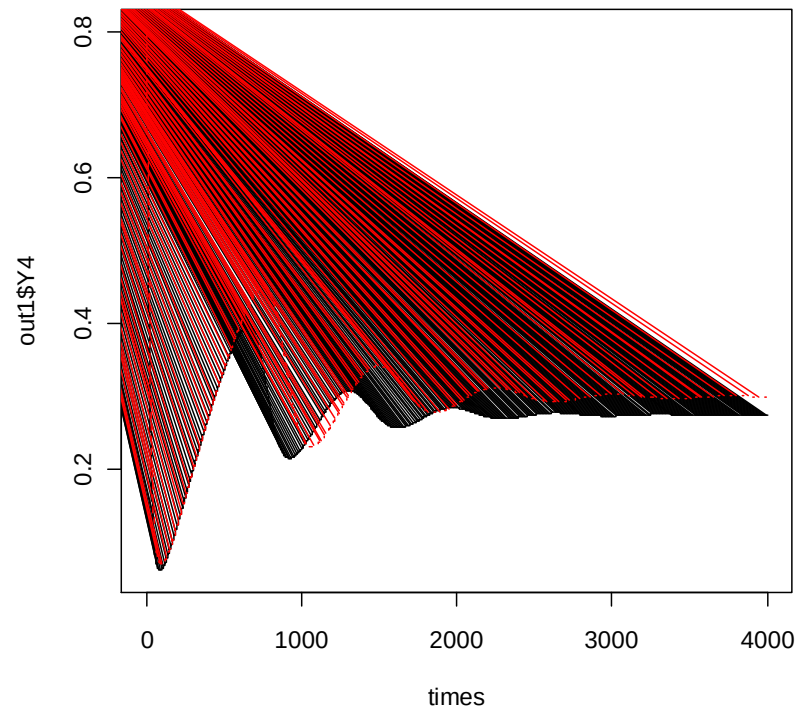
Transmission within / between the populations.

Susceptible in the two sub population

Susceptible in population 1



Susceptible in population 2



Part 4: age structured population

Interactive tutorial

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File C:/Ziv_Temp_2023/Tarylee_STEAM-3D/Workshop1_12_10_26/ShortCourse/Practicalessions_New/er_prog4_STEAM3D_2026-V1.html

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Code

10-09-2026 >eR-BioStat and STAEM-3D

Modeling infectious diseases using R - age structure population

Ziv Shkedy and Rashid Aloni based on Chapter 2-4 in the book *Modeling infectious diseases parameters based on serological and social contact data* (<https://www.imperial.ac.uk/micro-or/bio/stat/>)

Show

1. The SIR model with age structure population model

Contact pattern in age structure population

A special case of a SIR model with interacting subpopulations is an age-time dependent SIR model or age-structured SIR model. In such a model the population is divided into a finite number of age groups interacting with each other. Note that compared with the model discussed in the previous section, the flow of individuals into the susceptible class is possible only in the first age class. A second difference is that in the current model, individuals can pass to the same compartment in successive age classes, this means that a susceptible individual in age group i can remain susceptible and transfer (age passes) to the susceptible compartment in the successive age group, i.e., this individual passes from S_i to S_{i+1} .

The contact matrix

The transmission model

For a population with K age groups, the system of ordinary differential equations for the first age group ($i = 1$) in an age-structured SIR model is given by

$$\begin{cases} \frac{dS_i(t)}{dt} = - \left(\sum_{j=1}^K \beta_{ij} I_j \right) S_i + N \mu_i - \mu_i S_i - \eta_h S_i, \\ \frac{dI_i(t)}{dt} = \left(\sum_{j=1}^K \beta_{ij} I_j \right) S_i - \nu_i I_i - \eta_h I_i, \\ \frac{dR_i(t)}{dt} = \nu_i I_i - \mu_i R_i - \eta_h R_i. \end{cases}$$

Here, η_h is the rate at which individuals pass from S_i, I_i, R_i to $S_{i+1}, I_{i+1}, R_{i+1}$. From the second age group onwards, the system

Straks regen over ongeveer 1...

Search

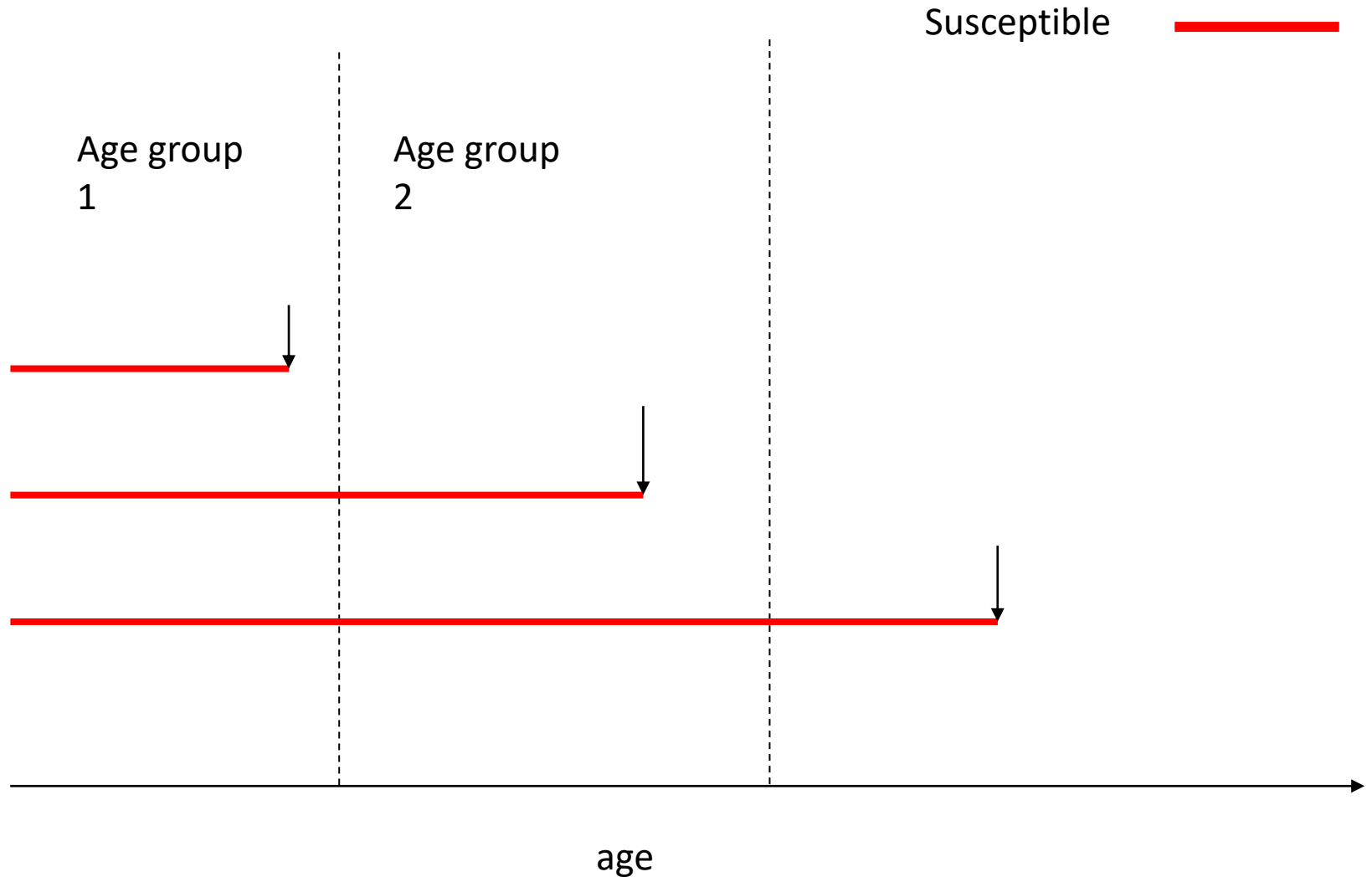
ENG 9:02 1/10/2026

Rmd program: eR_prog4_STEAM3D_2026_V2.Rmd

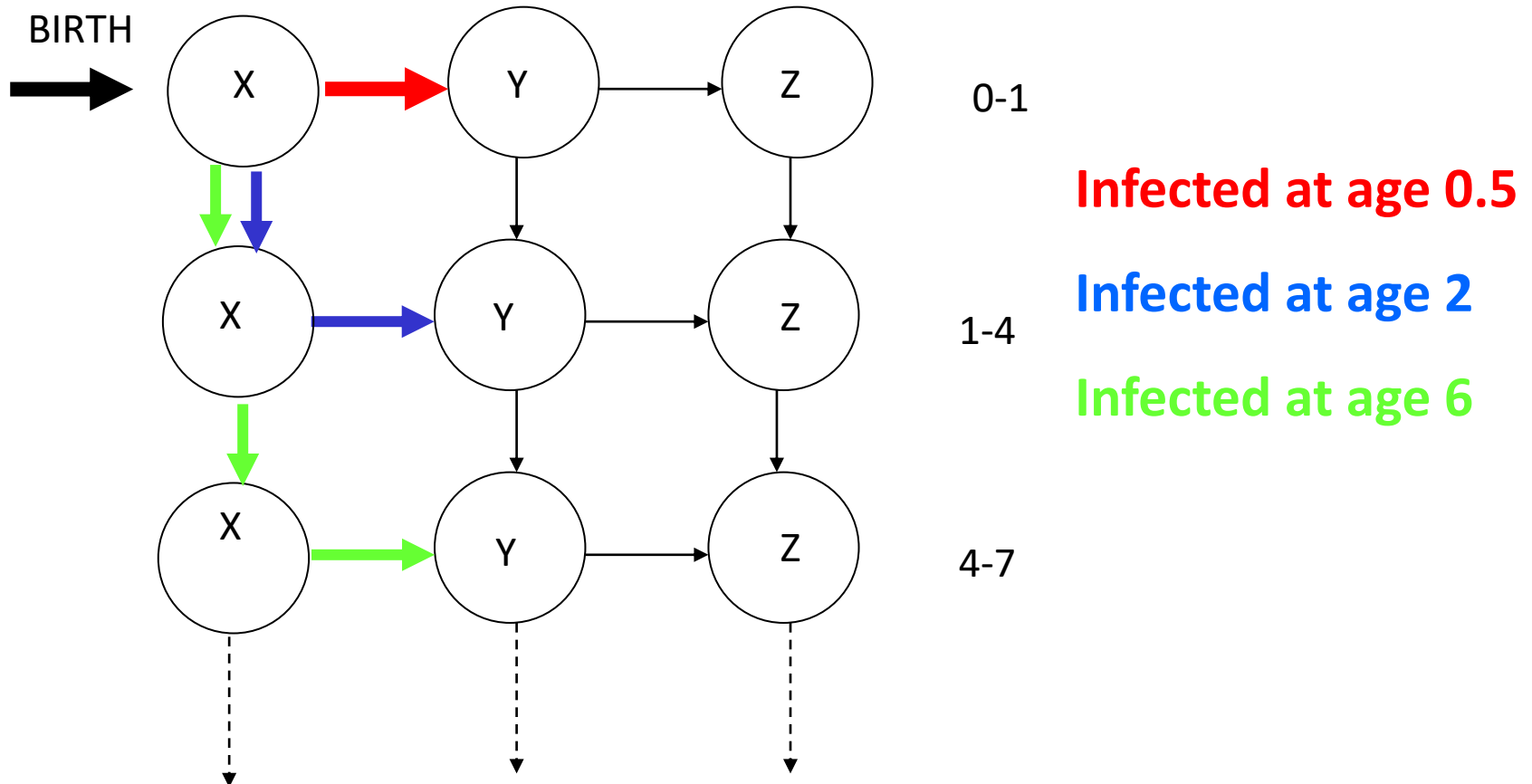
Age Structured population

- The population is divided by age groups.
- Examples:
- People has a tendency to mix with their own age group (children in the same class at school, students in the university).
- Children and parents (between groups transmission).

Age Structured population



Age Structured population



Contact patterns

Population with two age groups:

Case 1: transmission only
within the age groups

$$\begin{bmatrix} \beta_{11} & 0 \\ 0 & \beta_{22} \end{bmatrix}$$

Case 2: transmission
within and between the
age groups

$$\begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{12} & \beta_{22} \end{bmatrix}$$

Contact patterns

- Population with two age groups.
- Case 1: transmission only within the age groups

$$\begin{bmatrix} \beta_{11} & 0 \\ 0 & \beta_{22} \end{bmatrix}$$

```
parameters <- c(beta11=0.0001,beta12=0.00000,beta21=0.00000,  
  beta22=0.0001,v1=4,v2=4,mu=1/75,mu2=1/20,N=1000000)
```

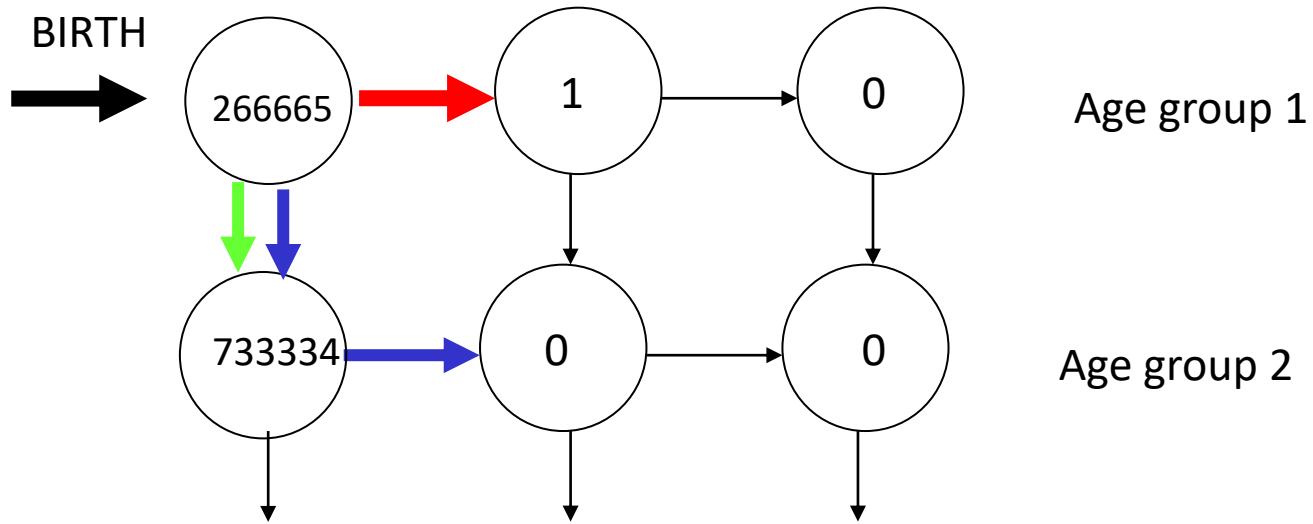
Contact patterns

- Population with two age groups.
- Case 2: transmission within and between the age groups:

$$\begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{12} & \beta_{22} \end{bmatrix}$$

```
parameters <-  
c(beta11=0.0001, beta12=0.0000075, beta21=0.0000075, beta22=0.0001,  
  v1=4, v2=4, mu=1/75, mu2=1/20, N=1000000)
```

State variables



```
state <- c(Y1=266665,Y2=1,Y3=0,Y4=733334,Y5=0.0,Y6=0)
```

The HTML file

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er_prog4_STEAM3D_2026-V2.html Open in Browser Find Publish

An age structure model in R

Definition of the model's parameters

For transmission rates $\beta_11 = \beta_22 = 0.0001$ and $\beta_12 = \beta_21 = 0.0000075$ we have

$$C = \begin{pmatrix} 0.0001 & 0.0000075 \\ 0.0000075 & 0.0001 \end{pmatrix}.$$

We assume a population with two age groups, 0-20 and 20-75, life expectancy of 75 years (birth rate=death rate= $\frac{1}{75}$) and a population size of 1000000 individuals. In R, the parameter vector is given by

Show

```
##      beta11      beta12      beta21      beta22      v1      v2
## 1.000000e-04 7.500000e-06 7.500000e-06 1.000000e-04 4.000000e+00 4.000000e+00
##           mu           mu2           N
## 1.333333e-02 5.000000e-02 1.000000e+06
```

In the first age group, $S(0) = 266665$, $I(0)=1$, $R(0)=0$ and in the second age group $S(0) = 733334$, $I(0)=0$, $R(0)=0$. The state vector in R is

The state vector

Show

```
##      Y1      Y2      Y3      Y4      Y5      Y6
## 266665      1      0 733334      0      0
```

Show

The transmission model

How to produce the HTML file ?

- Open the Rmd program: `eR_prog4_STEAM3D_2026_V2.Rmd`.
- Run the program.

Model Structure: two age groups

$$\begin{aligned}\frac{dS_1(t)}{dt} &= N\mu - (\beta_{11}I_1 + \beta_{12}I_2)S_1 - \mu S_1 - \eta S_1 \\ \frac{dI_1(t)}{dt} &= (\beta_{11}I_1 + \beta_{12}I_2)S_1 - \nu_1 I_1 - \mu I_1 - \eta I_1 \\ \frac{dR_1(t)}{dt} &= \nu_1 I_1 - \mu R_1 - \eta R_1 \\ \frac{dS_2(t)}{dt} &= \eta S_1 - (\beta_{21}I_1 + \beta_{22}I_2)S_2 - \mu S_2 \\ \frac{dI_2(t)}{dt} &= \eta I_1 + (\beta_{21}I_1 + \beta_{22}I_2)S_2 - \nu_2 I_2 - \mu I_2 \\ \frac{dR_2(t)}{dt} &= \eta R_1 + \nu_2 I_2 - \mu R_2\end{aligned}$$

```
SIRtwo<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dY1 <- -(beta11*Y2+beta12*Y5)*Y1+N*mu-mu*Y1-mu2*Y1
    dY2 <- (beta11*Y2+beta12*Y5)*Y1-v1*Y2-mu*Y2-mu2*Y2
    dY3 <- v1*Y2 - mu*Y3-mu2*Y3
    dY4 <- -(beta21*Y2+beta22*Y5)*Y4-mu*Y4+mu2*Y1
    dY5 <- (beta21*Y2+beta22*Y5)*Y4-v2*Y5-mu*Y5+mu2*Y2
    dY6 <- v2*Y5-mu*Y6+mu2*Y3
    list(c(dY1,dY2,dY3,dY4,dY5,dY6))
  })
}
```

$$\begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}$$

Move from the first age group to the second age group.

Parameter setting

$$N = 1000000$$

$$\mu_2 = \frac{1}{20} \quad \text{Age group: 0-20, 20-75}$$

$$\mu = \frac{1}{75} \quad \text{Life expectancy: 75 years}$$

$$v_1 = v_2 = 4$$

$$\begin{bmatrix} \beta_{11} = 0.0001 & \beta_{12} = 0.000075 \\ \beta_{12} = 0.000075 & \beta_{22} = 0.0001 \end{bmatrix}$$

```
> parameters <- c(beta11=0.0001,beta12=0.0000075,beta21=0.0000075,beta22=0.0001,  
+                  v1=4,v2=4,mu=1/75,mu2=1/20,N=1000000)
```

```
>parameters
```

beta11	beta12	beta21	beta22	v1	v2
1.000000e-04	7.500000e-06	7.500000e-06	1.000000e-04	4.000000e+00	4.000000e+00
mu	mu2	N			
1.333333e-02	5.000000e-02	1.000000e+06			

State variables

$$S_1(0) = 266665$$

$$I_1(0) = 1$$

$$R_1(0) = 0$$

$$S_2(0) = 733334$$

$$I_2(0) = 0$$

$$R_2(0) = 0$$

```
> state <- c(Y1=266665, Y2=1, Y3=0, Y4=733334, Y5=0.0, Y6=0)
```

```
> state
```

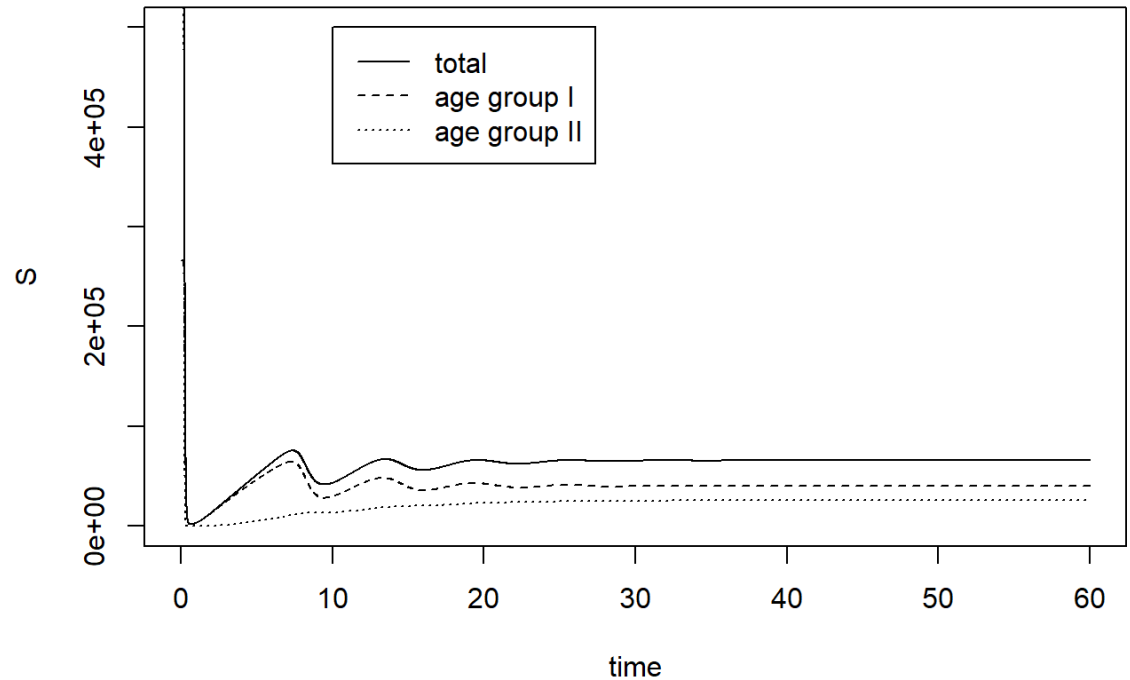
Y1	Y2	Y3	Y4	Y5	Y6
266665	1	0	733334	0	0

Model's output

Susceptible in the population

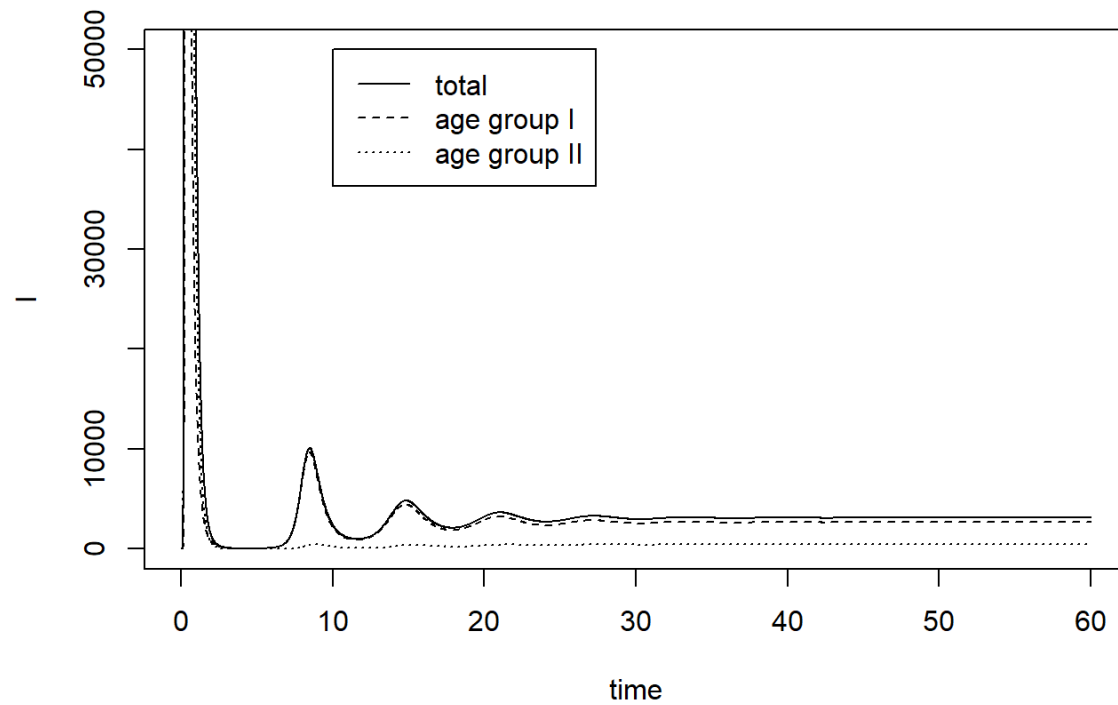
Parameter setting

$$\begin{bmatrix} \beta_{11} = 0.0001 & \beta_{12} = 0.000075 \\ \beta_{12} = 0.000075 & \beta_{22} = 0.0001 \end{bmatrix}$$



Model's output

- Infected in the population



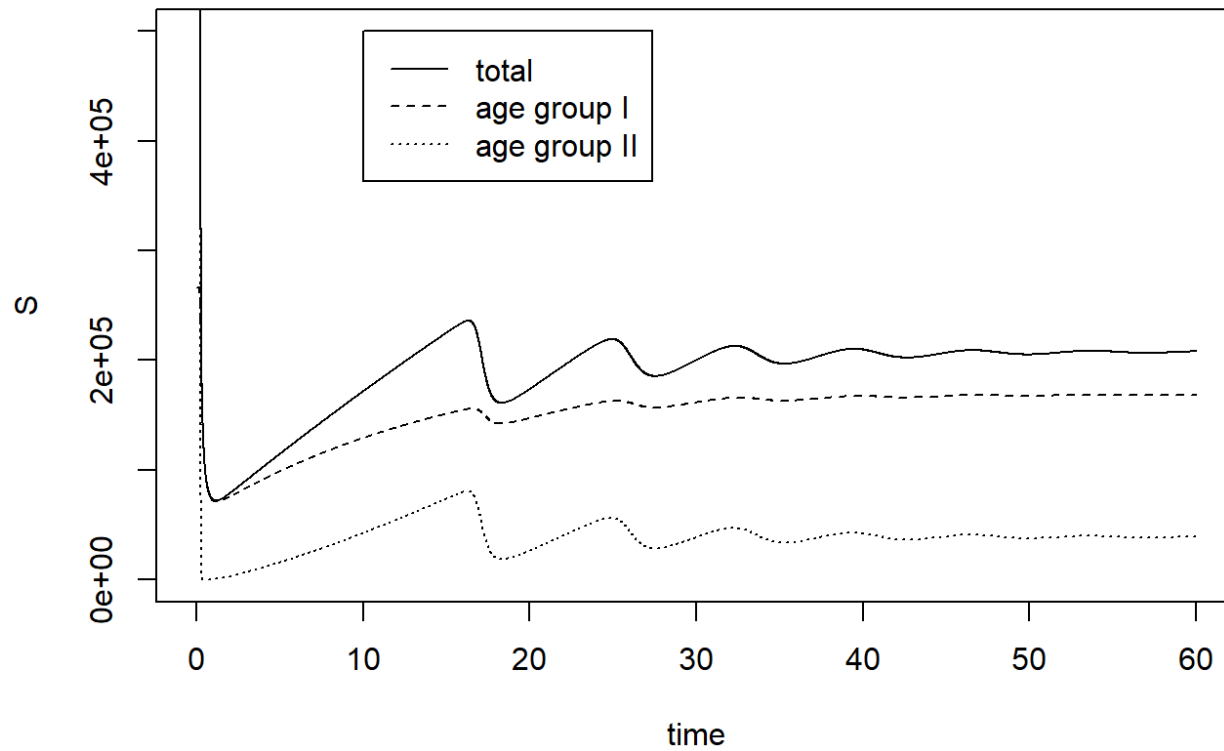
New parameter setting

- Population with two age groups.
- New parameter setting:


$$\begin{bmatrix} \beta_{11} = 0.000001 & \beta_{12} = 0.000075 \\ \beta_{12} = 0.000075 & \beta_{22} = 0.0001 \end{bmatrix}$$

- Lower transmission rate within the first age group.

Model's solution: susceptible



The HTML file

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1. The SIR model with age structure population model

2. Example I

Population structure with two age groups

An age structure model in R

Example 2: changing the parameter setting

time

Figure 2: SIR model with two age groups: infected.

Example 2: changing the parameter setting

A new contact matrix

Let us assume that the transmission rate within the first age group reduce from $\beta_{11} = 0.0001$ to $\beta_{11} = 0.000001$ while the other parameters remain constant. The new contact matrix is given by

$$C = \begin{pmatrix} 0.000001 & 0.0000075 \\ 0.0000075 & 0.0001 \end{pmatrix}.$$

In R, we updated the parameters vector

Show

```
##      beta11      beta12      beta21      beta22      v1      v2
## 1.000000e-06 7.500000e-06 7.500000e-06 1.000000e-04 4.000000e+00 4.000000e+00
##          mu      mu2      N
## 1.333333e-02 5.000000e-02 1.000000e+06
```

Running the model

Show

```
## time      Y1      Y2      Y3      Y4      Y5      Y6
## 1 0.00 266665.0 1.0000000 0.0000000 733334.0 0.0000000 0.0000000
## 2 0.01 266629.5 0.9634372 0.03924640 733369.5 0.07876116 0.001411269
## 3 0.02 266593.9 0.9304458 0.07707275 733404.8 0.23349479 0.007336380
## 4 0.03 266558.4 0.9030501 0.11365843 733440.0 0.54055317 0.022164270
## 5 0.04 266522.9 0.8853659 0.14930072 733474.8 1.15291406 0.054693785
## 6 0.05 266487.4 0.8856963 0.18453536 733509.0 2.37708411 0.122558953
```

Graphical output

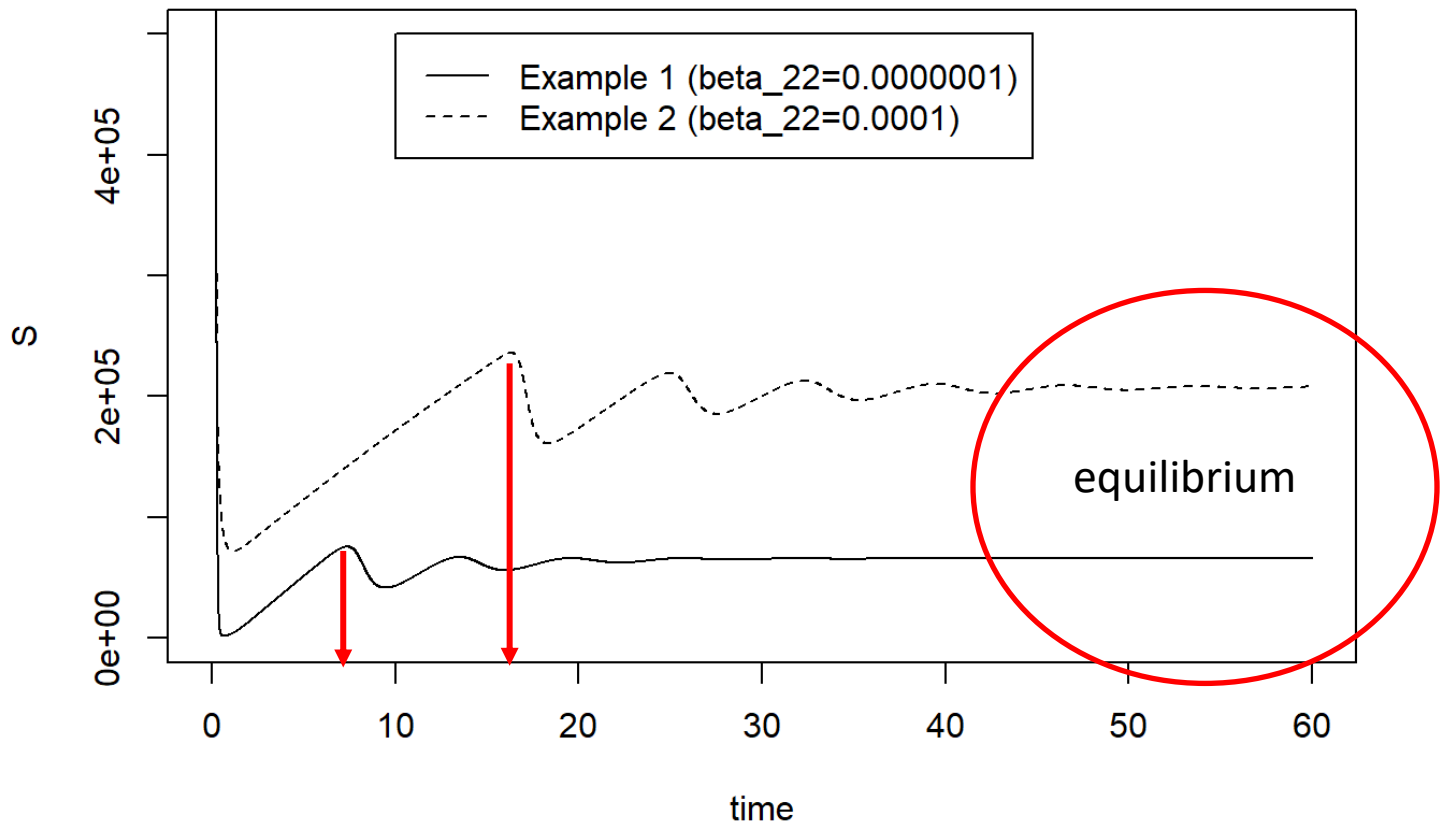
Show

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Model 's solution: susceptible



Model 's solution: infected

